Sand Dune

Design, Modeling, and Validation of a Background Oriented Schlieren System for High-Resolution Refractive Index Mapping of Transparent Media

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April 20, 2026

Core Principle

- A structured background is imaged *through* a transparent specimen
- Refractive-index gradients cause apparent pattern displacements
- Comparing reference and disturbed images yields the gradient field
- Integrating gradients reconstructs a relative map of Δn or Δt

Advantages

- Ideal for *weak* disturbances where interferometry is impractical
- Non-contact, full-field, and quantitative
- Sand Dune implements the *full* measurement chain, optical rig through reconstruction and export

Measurement Pipeline

1. Capture **reference** image (no sample)
2. Capture **flow** image (sample present)
3. Measure pixel **displacements** via cross-correlation
4. **Scale** displacements to physical gradient units
5. **Integrate** gradients to obtain RI or thickness map

Accuracy is dependent on the **entire chain**: background design, optical geometry, image quality, sub-pixel estimation, validation, and reconstruction.

Target Regime

Weak-to-moderate refractive disturbances, sub-pixel displacements sensitive to every stage.

Design Constraints

- Stable alignment: camera, specimen, background
- Repeatable sample positioning
- Background texture dense enough for reliable sub-pixel correlation

Key Geometric Parameters

Symbol	Description
Z_d	Background-to-sample distance
Z_a	Sample-to-lens distance
f	Lens focal length
P_{px}	Sensor pixel pitch

Hardware Components

- Camera + 25 mm fixed focal-length lens
- LED backlight panel (incoherent, no speckle)
- Blue optical filter (reduces chromatic aberration)
- Printed dot-pattern background targets (6 variants)
- Rigid support structure
- Custom specimen holder for repeatable positioning

A circular software mask restricts reconstruction to the sample aperture, excluding the holder.

System Design

Optical Layout & Hardware

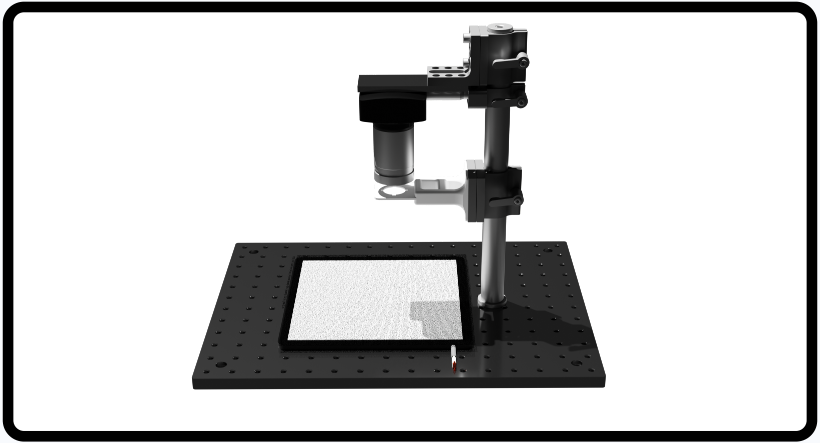


Figure: Hardware design of Sand Dune optical rig showing camera, lens, specimen region, and patterned background target.

Six Fabricated Patterns

- Fill densities: **40%** and **60%**
- Dot diameters: 0.10 mm, 0.50 mm, 0.75 mm

Optimal Dot Size: 4–5 px on sensor

Size	Effect
<3 px	Under-sampled; narrow noisy peak, poor CMM support
4–5 px	Well-resolved peak, broad CMM basin, maximum S2N
>7 px	Fewer features per window; lower S/N

Current optimum: 0.10 mm dots at 40% density.

Illumination

LED required. Coherent sources produce speckle, structured noise that degrades sub-pixel accuracy.

Focus

Out-of-focus dots blur edges and reduce contrast, degrading the correlation peak shape that CMM relies on. Focus the background as sharply as possible.

Density: 40% for large windows/coarse mag; 60% for small windows or near-lower-bound dot size.

BOS Measurement Model

Fundamental Quantity: Optical Path Length

$$\mathrm{OPL}(x,y) = \int_0^t n(x,y,z)\,dz \approx n(x,y) \cdot t$$

A lateral OPL gradient deflects the ray:

$$\varepsilon_x \approx \frac{t}{n}\frac{\partial n}{\partial x} \quad \Rightarrow \quad u_{px} = \frac{\varepsilon_x Z_d}{P_{px}} \cdot \frac{z_i}{Z_B}$$

Two Modes (one quantity must be known):

- **dn mode:** t known $\rightarrow$ reconstructs $\Delta n(x,y)$
- **mm mode:** n known $\rightarrow$ reconstructs $\Delta t(x,y)$

Fundamental Limitation

BOS measures integrated OPL and *cannot* independently separate n and t without additional information (profilometry, OCT, etc.).

Other Physical Contributors

- Flat-surface refraction offset (Snell's law, correctable)
- Sample tilt, uniform plane in u/v maps
- Surface curvature and figure errors
- Stress birefringence (scalar mean only)

All contributions are integrated into the surface map. Attribution requires independent measurements.

BOS Measurement Model

Image distance (thin-lens, $Z_B = Z_a + Z_d$):

$$z_i = \frac{f\,Z_B}{Z_B - f}$$

Correlation-map scale factor (dn mode):

$$S_{\text{corr}} = P_{px}\,\frac{n\,(Z_B - f)}{f\,Z_d\,t} \;\Rightarrow\; \frac{\partial n}{\partial x} = u_{px} \cdot S_{\text{corr}}$$

Sample-plane grid spacing:

$$\Delta x_{\text{grid}} = \texttt{step} \cdot P_{px} \cdot \frac{Z_a}{z_i}$$

Surface scale factor:

$$S_{\text{surf}} = S_{\text{corr}} \times \Delta x_{\text{grid}}$$

Optional Refraction Correction (Optional)

Off-axis rays through a flat slab shift the apparent background position even for a *uniform* sample.

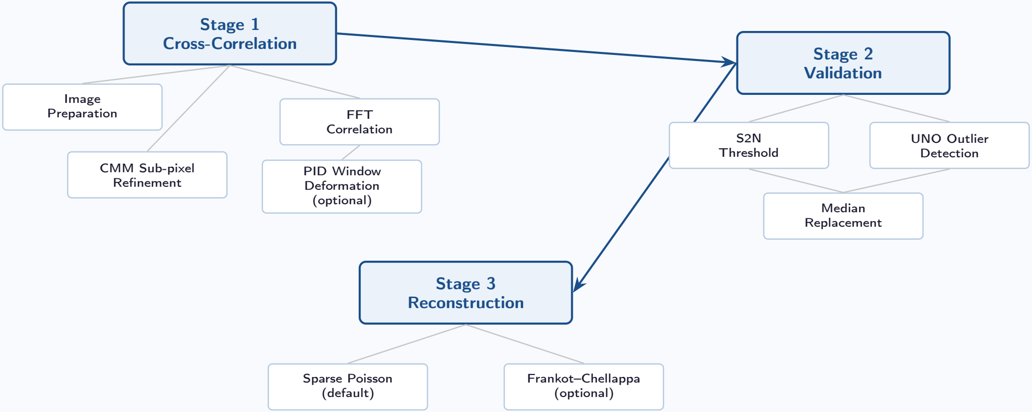
Computed via Snell's law and subtracted before scaling:

$$\theta_{r,x} = \arcsin\!\left(\frac{\sin\theta_x}{n}\right), \quad d_x = \frac{\sin(\theta_x - \theta_{r,x})}{\cos\theta_{r,x}}\,t$$

Significant for large n or thick t; negligible near $n \approx 1$.

Processing Pipeline

Three-Stage Overview



Window Grid (W = window size, O = overlap)

$$\texttt{step} = W - O \quad n_c = \left\lfloor \frac{W_{\text{img}} - W}{\texttt{step}} \right\rfloor + 1$$

Per-Window Steps

1. Extract reference and flow interrogation blocks
2. Skip if local variance $< 10^{-3}$ (flat patch)
3. Subtract local mean
4. Forward FFT of both windows
5. Cross-power spectrum $G = \overline{F_{\text{ref}}}\, F_{\text{flow}}$
6. Inverse FFT $\rightarrow$ correlation map ϕ; shift to centre

$$\phi = \mathcal{F}^{-1}\{\overline{F_{\text{ref}}}\, F_{\text{flow}}\}$$

Signal-to-Noise Ratio

$$\text{s2n} = \frac{\phi_1}{\phi_2}$$

ϕ_1: primary peak; ϕ_2: highest value outside $\pm 2\,\text{px}$ exclusion zone.

FFTW Implementation

FFTW3 (single-precision float) plans allocated once at `Correlator` construction and reused across all windows, avoiding per-frame planning overhead.

Processing Pipeline

Stage 1 — CMM Sub-Pixel Refinement

Correlation Mapping Method (Chen & Katz, 2005)

- 5×5 patch centred on the discrete correlation peak
- Minimises mean-normalised residual against the autocorrelation:

$$\varepsilon = \sum_{i=0}^{24} \left(\tilde{\phi}[i] - \tilde{\phi}'[i] \right)^2$$

- $\tilde{\phi}'$ evaluated at offset (u', v') via bicubic interpolation

Solver Sequence

1. **Seed:** log-parabola fit along dominant axis
2. **Gauss–Newton:** ≤ 8 iters, backtracking line-search
3. **Grid fallback:** dense $0.01\,\text{px}$ search if G–N fails
4. **Boundary handoff:** test adjacent cell if within $0.02\,\text{px}$ of ± 0.5

Solution clamped to $|u'|, |v'| \leq 0.49\,\text{px}$.

Peak Locking Suppression

Centroid/parabola fits are biased toward integer values, producing artefact spikes in displacement histograms.

CMM fits the *entire patch shape*, not just the peak tip, suppressing this bias. Boundary-handoff and the ± 0.49 clamp are additional safe-guards.

Processing Pipeline

Stage 1 — CMM Sub-Pixel Refinement

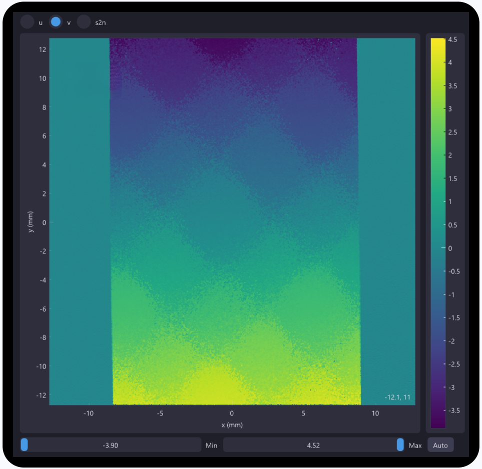


Figure: Peak locking when using gaussian interpolation for sub-pixel displacement.

Motivation: rigid windows ignore local shear and strain, biasing the displacement estimate for non-uniform fields.

Algorithm (per PID pass)

1. Compute 1st-order displacement gradients via central differences
2. Apply 3×3 box-filter smoothing (`pid_smoothing_passes`)
3. Build affine warp at each window centre
4. Resample flow window; re-correlate for residual
5. Accept update only if CMM residual improves
6. Repeat; stop early if mean correction $< 10^{-3}$ px

Parameters

`enable_pid`	on/off
`pid_iterations`	# PID passes
`pid_relaxation`	damping $\in [0, 1]$
`pid_smoothing_passes`	filter count

Measured Benefit (affine test field)

Method	RMSE (px)
Baseline (no PID)	0.2745
PID - 2 iters	0.0875
PID - 3 iters	0.0846

2 passes give **68**% RMSE reduction; 3rd adds only 3%.

Processing Pipeline

Pass 1: Signal-to-Noise Threshold

$$\text{Flag if} \quad \mathrm{s2n} < 1.3$$

Pass 2: Universal Normalized Residual (UNO) (Westerweel & Scarano, 2005)

$$\tilde{u}_{\mathrm{nrm}} = \frac{|u - u_{\mathrm{med}}|}{\mathrm{median}(|u_k - u_{\mathrm{med}}|) + 0.1}$$

$$\text{Flag if} \quad \sqrt{\tilde{u}^2_{\mathrm{nrm}} + \tilde{v}^2_{\mathrm{nrm}}} > 2.0$$

u_{med}: 8-connected neighbourhood median.

Replacement: flagged vectors → 8-neighbourhood median of unflagged neighbours.

UNO Purpose

The normalised residual adapts to local displacement magnitude, unlike a fixed absolute threshold, it remains effective across spatially varying signal levels.

Processing Pipeline

Default: Masked Sparse Poisson Solver

Minimise edge-gradient energy over masked valid nodes:

$$E(s) = \sum_{\text{x-edges}} \left(s_{i,j+1} - s_{i,j} - g^x_{i,j}\right)^2 + \sum_{\text{y-edges}} \left(s_{i+1,j} - s_{i,j} - g^y_{i,j}\right)^2$$

where $g^x_{i,j} = \frac{1}{2}(u_{i,j} + u_{i,j+1})$. Setting $\partial E / \partial s_{i,j} = 0$ yields a sparse system solved with `Eigen::ConjugateGradient`. First valid node pinned to zero.

Frankot–Chellappa (Optional)

FFT-based frequency-domain integration:

$$\hat{s} = \frac{i f_x \hat{u} + i f_y \hat{v}}{2\pi(f_x^2 + f_y^2) + \varepsilon}$$

Much faster but zeroing gradients outside the mask can introduce boundary artifacts on heavy masks.

Recommendation: Sparse Poisson for masked datasets; Frankot–Chellappa for fast comparison on full-domain fields.

Software Architecture
Build System & Key Classes

Build: CMake, C++23. Targets: `sand_dune` (GUI) and `run_tests` (test binary).

Library	Role
SDL3, SDL3_image	Window, render, image I/O
Dear ImGui	All UI panels & controls
ImPlot	Heatmap visualisations
Eigen3	Dense/sparse linear algebra
FFTW3 (float)	FFT correlation & FC
doctest	Unit-test framework

Key Classes

- `Session`: orchestrator; owns images, intermediate arrays, parameter structs, and async stage-state machine (`std::atomic<StageState>`)
- `Correlator`: reference + flow $\rightarrow$ `VectorField`; FFTW plans allocated once at construction
- `Reconstruction`: sparse Poisson and Frankot–Chellappa back ends
- `VectorField`: stores u, v, s2n matrices; exposes `SaveCSV()`
- `Image`: normalized $[0, 1]$ float matrix loader

Stages run via `std::async`; UI polls stage states to update progress.

Using the Application | Typical Workflow

1. **Load images**: Reference + one or more flow images via the **Load** panel. All images must share identical pixel dimensions.
2. **Set parameters**: Optical geometry (Z_d, Z_a, f, P_{px}), sample properties (n, t), correlation settings. The **Calculations** panel updates live with derived grid spacing, physical field width, and feature size estimate.
3. **Configure the mask**: Enable circular mask; set centre and radius to enclose the sample region.
4. **Run pipeline**: **Run Full Pipeline** (all three stages) or each stage individually. Progress bar and ETA shown per stage.
5. **Inspect results**: Correlation, Validation, and Surface tabs; toggle raw pixel / physical units; adjust colormap via min/max sliders.
6. **Export**: Choose output directory in the **Save** panel. Three CSV files per frame: `_correlation.csv`, `_val.csv`, `_surface.csv`.

Using the Application

Parameters Panel Reference

Group	Control	Effect
Correlation	Window size	Interrogation window side length (px)
	Overlap	Window overlap; $\texttt{step} = W - O$
	Enable PID	Iterative window deformation
	PID iterations	Number of passes after baseline
	PID relaxation	Damping on residual updates $[0, 1]$
Optical	P_{px}	Sensor pixel pitch (µm)
	Z_d, Z_a, f	Geometry distances (mm)
Sample	n, t	Refractive index and thickness (mm)
Reconstruction	Mode	RI variation (dn) or thickness variation (mm)
	Apply correction	Subtract Snell's-law geometric offset
Mask	Enable / Centre / Radius	Restrict reconstruction to circular ROI

Validation & Precision

Image-Based Precision Metric

Test Image Pair: `if_0.1`
Glass window of *known uniform* RI and thickness.
Expected field: smooth, nearly planar, plane-fit residual quantifies spurious structure (striping, locking artifacts).

Metric	u (px)	v (px)
Whole-map, no PID	0.659	0.670
Whole-map, PID	0.596	0.579
Central crop, no PID	0.096	0.106
Central crop, PID	**0.049**	**0.046**

Central crop = inner 80%; outer 10% per side excluded (edge artifacts).

PID Improvement on Central Crop

- $\approx$ **49.5%** reduction in u-RMS
- $\approx$ **56.8%** reduction in v-RMS

Additional Diagnostics

Striping: for `if_0.1`, dominant-axis correlation was 0.9985 (u) and 0.9981 (v); orthogonal-axis correlation was 0.0052 and 0.0068.

Peak-locking: half-pixel locked fraction was 0.0430/0.0436 raw and 0.0425/0.0431 validated for u/v.

Residual ratio: decreased from $0.0517 \rightarrow 0.0505$ in u and $0.0554 \rightarrow 0.0538$ in v after validation.

Validation & Precision | Image-Based Precision Metric

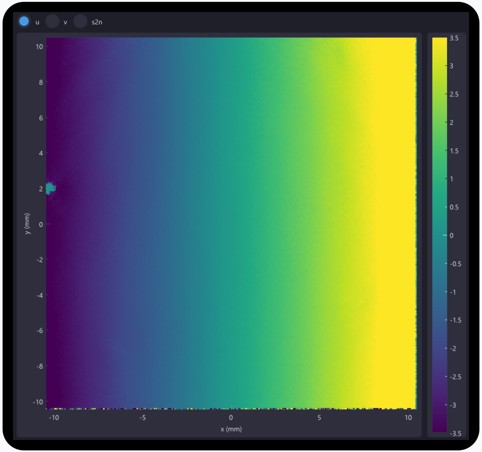


Figure: Optical window measured using BOS system.

System Precision Estimate | Minimum Resolvable Gradient

Current Optical Configuration

Parameter	Symbol	Value
Focal length	f	25 mm
BG–sample dist.	Z_d	220 mm
Sample–lens dist.	Z_a	45 mm
Total obj. dist.	Z_B	265 mm
Pixel pitch	P_{px}	2.315 µm
Sample RI	n	1.08
Thickness	t	0.924 mm

Min. detectable displacement (3σ):

$$u_{\min} = 3 \times 0.049 \approx 0.15 \text{ px}$$

$$S_{\text{corr}} \approx 0.1181 \text{ m}^{-1}\text{px}^{-1}$$

Min. resolvable gradient:

$$\left.\frac{dn}{dx}\right|_{\min} \approx 1.77 \times 10^{-5} \text{ dn/mm}$$

Worst-case expected (Δn=0.005 over 24 mm):

$$\left.\frac{dn}{dx}\right|_{\text{wc}} \approx 2.08 \times 10^{-4} \text{ dn/mm}$$

Detection margin $\approx$ **11.8**$\times$

Even in the shallowest expected gradient, the system resolves the signal by $\sim$one order of magnitude.

Calibration & Interpretation

Calibrating Z_a
Most sensitive parameter: an error in Z_a shifts *both* the displayed field width and the reconstructed surface amplitude (both depend on Δx_{grid}).

Do not use a ruler directly. Recommended procedure:

1. Place a target of **known physical width** in the background plane
2. Record how many pixels span that width
3. Back-calculate Z_a until the **Calculations** panel field width matches the known dimension

The reconstructed surface has an arbitrary constant offset; absolute comparisons across experiments require an independent reference point.

Reconstruction vs. Gradient Maps

Use the surface map to:
Characterise overall uniformity; quantify large-scale RI or thickness variation; compare samples.

Use the u/v gradient maps to:
Locate localised defects, inclusions, bubbles, cracks, steep-gradient regions, that may be smeared or suppressed by the integration step.

Always inspect both together.

Known Caveats

Fundamental & Software Limitations

Measurement Fundamentals

- n **and** t **cannot be decoupled** from BOS data alone
- **Not a pure RI map**: tilt, curvature, birefringence all contribute
- **Line-of-sight integrated**: 3D depth structure not resolved
- $\partial n / \partial z$ **invisible**: only transverse gradients deflect rays
- **Gradient, not a surface**: integration accumulates errors; inspect both maps
- **Small-angle approx**: breaks down for large deflections
- **Reference must be distortion-free**: lens distortion appears as a systematic offset

Software & Implementation

- **Curl-free assumption**: Poisson solver discards non-integrable curl silently; high-curl field can look deceptively smooth
- **FC mask treatment** weaker than sparse Poisson for masked domains
- **Surface is relative**: gauge constant is arbitrary
- **Geometry calibration critical**: incorrect Z_a shifts both field width and amplitude by the same factor

Conclusion

Sand Dune is a complete BOS measurement system for mapping weak optical non-uniformities in transparent media at high spatial resolution.

Key Capabilities

- FFT cross-correlation + CMM sub-pixel refinement
- Optional PID window deformation ($\sim 50\%$ precision improvement)
- UNO-based two-pass vector validation
- Masked sparse Poisson (default) and Frankot–Chellappa (alternative)
- Real-time parameter calculations and interactive heatmap visualisation
- CSV export for all pipeline stages

Current Precision Benchmark
Central-crop plane-fit RMS with PID:
$\approx 0.049\,\mathrm{px}$ (u) and $0.046\,\mathrm{px}$ (v)

Detection Margin

$11.8\times$ margin over worst-case gradient.

Refs: Raffel (2015) · Chen & Katz (2005) · Westerweel & Scarano (2005) · Frankot & Chellappa (1988) · Agrawal et al. (2006) · Liu et al. (2022)

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